Mirage PUF: FPGA Implementation and Machine Learning Resistance Characterization

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Abstract
The growing capability of AI has transformed machine learning attacks on Physical Unclonable Functions (PUFs) from a theoretical concern into a practical threat. Architectures once considered secure—including XOR, feed-forward, and interpose Arbiter PUFs—have been broken by gradient boosting and deep neural networks. This work presents the Mirage PUF, a structural hardware countermeasure that eliminates ML learnability at the architectural level. Rather than relying on standard FPGA routing, it uses CARRY4 primitives to create symmetric delay paths. A secondary carry chain, tapped at four depths, captures an unobservable transient state that generates a dynamic avalanche mask. This mutates the primary challenge during evaluation while remaining entirely internal to the FPGA, making the true challenge permanently inaccessible. Across nearly 25,000 challenge-response pairs, Mirage PUF achieves 49.99% inter-challenge randomness. Under a white-box threat model with full architectural knowledge, gradient boosting and deep neural networks achieve only 56.20% prediction accuracy, effectively random, demonstrating strong resistance against AI-driven modeling attacks.

Keywords: CARRY4, FPGA, Hardware Security, Machine Learning Attacks, Physical Unclonable Functions.

I. Introduction

Authentication is a persistent problem with IoT deployments. Traditional security approaches store static secret keys on non-volatile memory (NVM). However, this model is rather vulnerable against attackers with physical access to the device. Physical Unclonable Functions (PUFs) offer an alternative by generating a key on demand from the microscopic manufacturing variations unique to each silicon instance instead of storing it. This makes authentication much more secure against physical attacks.

Among intrinsic PUF designs, delay-based architectures—particularly the Arbiter PUF (APUF) and Carry-Chain Delay Line (CCDL) PUF—are widely adopted on FPGAs for their compact footprint [1]. Designs such as the Anderson PUF [2] demonstrated the concept but proved difficult to scale on modern FPGAs, where the fixed routing fabric [3]

introduces deterministic interconnect skew that disrupts path symmetry. The skew can be offset by PDL compensation and manual macro-placement, but produce device-specific calibrations that do not transfer across dies.

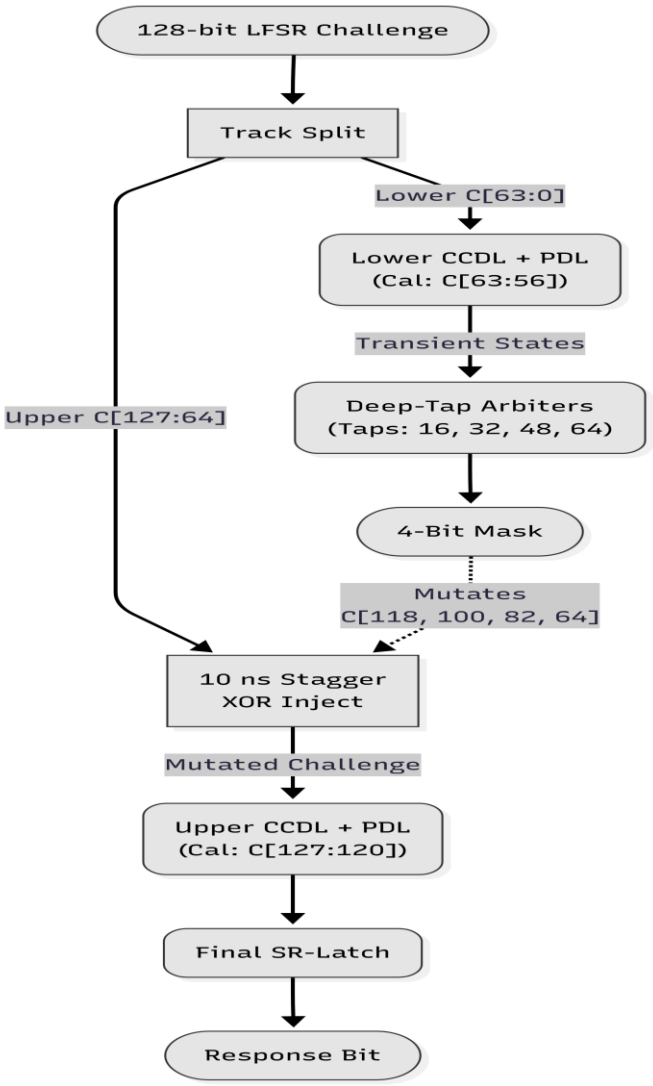
Moreover, since the challenge-response relationship works on a linearly additive delay model, they are highly susceptible to attacks by ML models trained on a sufficient number of challenge-response pairs (CRPs) [4, 5, 6]. Advanced architectures like the XOR-APUF, the Feed-Forward APUF [7], and the Interpose PUF [8] introduce non-linearity by chaining or interposing multiple arbiters, but remain vulnerable to Gradient Boosting and Deep Neural Network estimators [9]. Environmental fluctuations further degrade reliability [10, 11]. The standard solution of on-chip BCH error correction [12] imposes area and power overheads.

The Avalanche-Masked architecture (Mirage PUF) addresses ML vulnerability by linking the primary delay path to the unobservable analog state of a secondary carry-chain mid-evaluation. This forces the primary path to dynamically change based on unobservable internal chip behavior. The principal contributions are as follows:

- **Mid-Flight Masking for ML Resistance:** The design captures intermediate response bits from a secondary PUF, generating a 4-bit mask injected into the primary PUF's challenge vector mid-evaluation. This restricts advanced ML estimators—including gradient boosting and deep neural networks—to a 56.20% prediction ceiling across 25,000-CRP datasets. The design only consumes a core footprint of 32 LUTs and 64 CARRY4 primitives.
- **Inter-Challenge Hamming Distance as a Metric to Measure Randomness:** Inter-Challenge HD is used to measure how much the response changes for different challenges on the same chip. This addresses a gap in the APUF literature where intra-device avalanche behavior is frequently conflated with inter-device spatial uniqueness. The Mirage PUF achieves 49.99% Inter-Challenge HD, proving the randomized behavior.
- **Physical Entropy Limits of CARRY4 Routing:** Systematic evaluation across multiple Artix-7 FPGA instances demonstrates that CARRY4-exclusive delay architectures exhibit a severe Inter-Die HD of 13.04%. This happens because the routing in these carry chains is very fixed and predictable, which reduces natural variation. So, auxiliary entropy sources are needed to make each chip truly unique.

The remainder of this article is organized as follows. Section II describes the Mirage PUF hardware architecture. Section III outlines the experimental setup and metric definitions. Section IV presents empirical results. Section V concludes.

II. The Mirage PUF Architecture



[Figure 1: Schematic of the Mirage PUF architecture. The deep-tap mask from the lower track dynamically mutates the upper routing path mid-evaluation.]

A. Design Evolution and the Linearity Constraint

The Mirage PUF topology was developed by improving on three successive FPGA-based PUF architectures, each addressing the primary failure of its predecessor.

Anderson PUFs use LUT-based delay chains routed through the general-purpose FPGA fabric. On modern devices, automated routing guarantees uncontrollable path asymmetry, causing uneven delays. A successful model requires exhaustive manual routing on every single board separately, limiting reproducibility.

CCDL (Carry Chain Delay Logic) PUFs substitute LUT chains with dedicated vertical Xilinx CARRY4 columns. Routing across CARRY4 primitives bypasses the general-

purpose FPGA routing matrix entirely, resulting in a highly reproducible architecture. However, the PUF is still strictly additive. This results in the challenge-response relation being highly linear, leaving the architecture still vulnerable to ML attacks.

XOR-CCDL PUFs chain multiple CCDL PUFs and XOR their outputs, introducing non-linearity. However, this non-linearity, while enough to resist weaker ML models, is not nearly enough to protect against Deep Neural Networks and Gradient Boosting frameworks. These advanced ML models can still predict the response for a challenge with considerable accuracy given enough CRPs [8]. XORing more PUFs together can increase security, but also takes up much more of the FPGA's area.

B. Mid-Flight Mutation via Unobservable Silicon States

It is now clear that obfuscating the output of a linear delay chain does not remove the underlying linearity from the challenge response relationship. True resistance against ML modeling attacks requires a different approach. The Mirage PUF architecture takes the approach of altering the challenge during evaluation using variables that are physically inaccessible to any external observer.

The Mirage PUF achieves this with a secondary CCDL PUF that functions not as a key generator but as a hardware mutator. The analog transient state of this secondary chain—shaped by nanoscale process variation of its specific silicon instance—is captured at multiple depths and used to dynamically modify the challenge presented to the primary chain before propagation completes.

C. The CARRY4 Primitive and Structural Determinism

The core delay elements of the Mirage PUF are constructed using the dedicated CARRY4 primitives inherent to the Xilinx 7-series architecture. Each CARRY4 block is subdivided into four internal stages (representing bits 0 through 3), with each stage containing a dedicated 2-to-1 multiplexer (MUXCY) and an XOR gate. Unlike standard delay lines that utilize Look-Up Tables (LUTs) connected via the programmable switch matrix, CARRY4 columns utilize hardwired traces. At the micro-level, a single internal stage i is governed by a strict port mapping: it accepts a local data input (DI[i]) and a multiplexer select line (S[i]), producing an arithmetic sum (O[i]) and a localized carry-out (CO[i]). Internally, the carry-out of one stage feeds directly into the multiplexer of the next. To construct the continuous highway, these routing traces bypass the general interconnect entirely, cascading the final carry-out bit of the top stage (CO[3]) directly into the cascade carry-in (CI) of the adjacent CARRY4 primitive in the FPGA slice above it.

Furthermore, the internal MUXCY multiplexers act as the engine that allows the Mirage PUF to generate a full 64-bit response block from a single challenge. The multiplexer can be controlled via its Select (S) pin, which is connected to a specific challenge bit, to force the signal to propagate through either a slower (DI) or a faster (CI) path at every stage.

This is the basis of strong PUFs, where each unique challenge creates a unique path and generates a distinct response. As a result, a large number of CRPs can be realized. Therefore, while each challenge only gets a single-bit response, it can be simply shifted repeatedly to get new paths each time, concatenating each 1-bit response to form a complete key.

This architecture provides a critical advantage for scalable deployment and mass manufacturing. Standard LUT-based PUFs suffer from uncontrollable placement and routing (P&R) variations, often requiring exhaustive, per-device manual routing interventions to maintain path symmetry. By confining the main PUF tracks strictly to the hardwired CARRY4 columns, the Mirage PUF boasts high reproducibility across any Xilinx 7-series FPGA. This enables automation for mass production without the need for localized routing constraints.

D. Hardware Implementation

The architecture consists of two parallel tracks: a lower Deep-Tap mutator and an upper primary delay line.

The lower chain is tapped at the 1/4, 2/4, 3/4, and 4/4 positions along its column. At each tap, a dedicated SR-latch arbiter samples the propagating signal. SR-latches are selected over clocked flip-flops for their sensitivity to sub-threshold analog signal levels: a clocked element resolves metastable intermediate states by waiting for the clock edge, whereas the SR-latch captures the analog balance between set and reset paths at the moment of sampling, encoding finer-grained silicon variation into the output bit.

The four captured bits form the avalanche mask, which is XOR-injected into the S-pin routing multiplexers of the upper CARRY4 chain at challenge bit positions 0, 18, 36, and 54. An asynchronous stagger constrained to slice X36 delays the upper chain's evaluation by ~10 ns, ensuring the lower avalanche mask has fully resolved before the upper chain begins propagating, preventing race conditions at the injection points.

E. Cryptographic Non-Linearity of the Avalanche Mechanism

A single-bit change in the input challenge alters propagation timing in the lower chain, which changes the balance at one or more SR-latch arbiters, producing a different mask configuration. This mask is XOR-injected at four positions in the upper chain, reshaping the routing permutation of the upper CARRY4 multiplexers across the entire column. Because the mask is generated by analog silicon states that no external observer can measure or reproduce from the CRP dataset alone, no gradient-based estimator can learn the true challenge-to-response mapping. The upper chain's effective input is never directly observable—only the final response is—making the architecture structurally opaque to the ML attack classes evaluated in Section IV.

F. Data Acquisition and UART Interface

A hardware wrapper was synthesized alongside the Mirage PUF core for automated CRP extraction. A segmented LFSR on-chip generated challenge vectors by sweeping the lower 56 bits of the 128-bit challenge bus, while the upper 8 bits (C_63:56 and C_127:120) were held fixed at the calibrated PDL coordinates. Although 16 bits are locked for calibration, the remaining 112 bits yield an effective challenge space of 2^112 permutations—cryptographically vast and practically inexhaustible for brute-force profiling. The LFSR's pseudorandom stimulus does not compromise the Inter-Challenge HD measurement: the 49.99% decorrelation reflects the non-linearity of the avalanche mask, not the LFSR's own bit distribution, as evidenced by the failure of all tested ML models to extract a learnable structure from the dataset.

An integrated FSM transmitted 64-bit response blocks over a 115,200 bps UART link to a host machine, enabling fully automated extraction of the 25,000-CRP dataset per board.

III. Experimental Setup

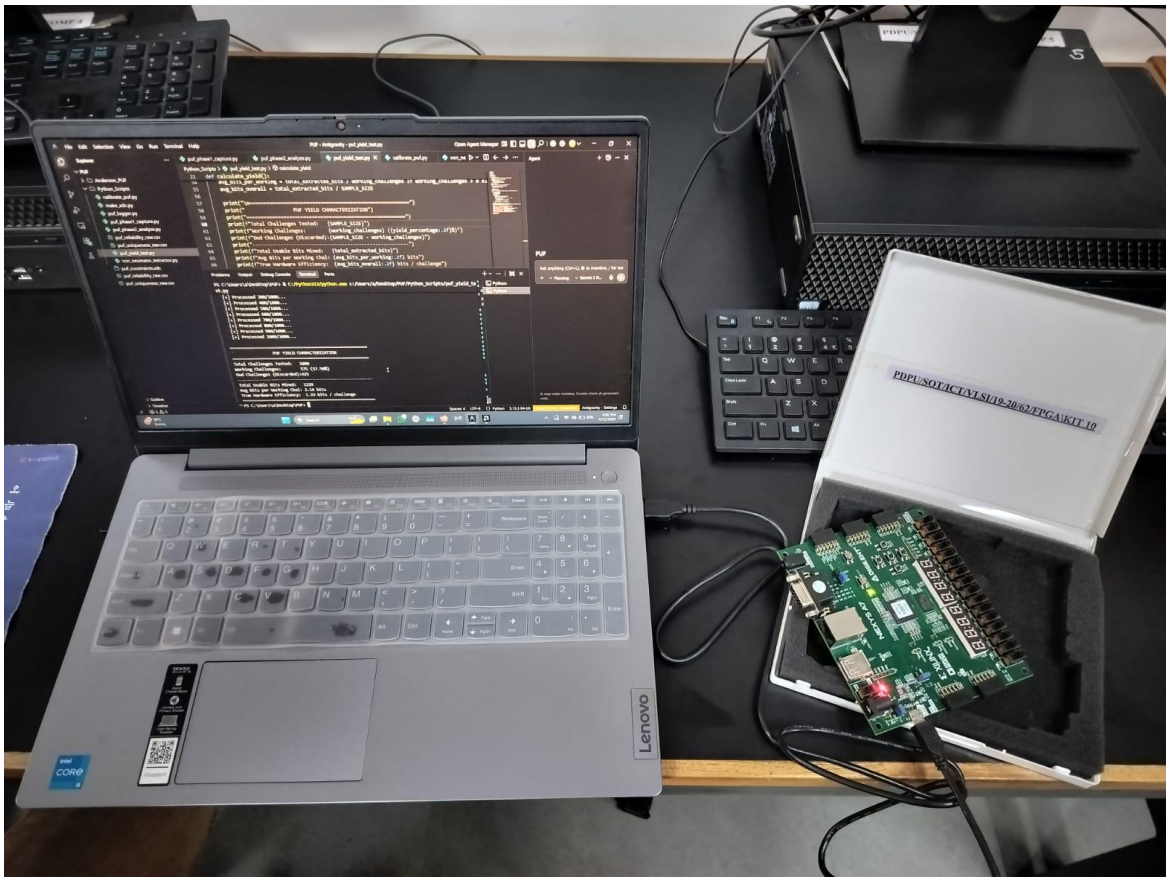


Fig. 1. Physical experimental setup: Nexys A7-100T FPGA board connected to host machine via UART for automated CRP extraction.

A. Hardware Platform and Operating Conditions

All evaluations were conducted on Artix-7 XC7A100T FPGAs (Nexys A7-100T) boards, operated at a nominal 1.0 V core voltage under ambient laboratory conditions (≈25 °C) without active thermal regulation.

B. Design Space Exploration and PDL Calibration

To assess whether spatial placement could recover Inter-Die HD, the Mirage PUF macro was instantiated across multiple distinct Pblocks and CARRY4 columns using custom Tcl scripts. Despite exhaustive placement variation, multi-chip signatures remained highly correlated—confirming that CARRY4 routing determinism is a device-family property, not a placement artifact. A PDL calibration sweep over the upper 8 challenge bits identified the operating point with the lowest Intra-HD, ensuring stable CRP extraction.

C. CRP Extraction Protocol

The primary board dataset (25,000 challenges, 10 evaluations each, yielding 250,000 raw readings) was used for analyzing intra-device reliability and randomness, and CRP space stability. Supplementary boards were evaluated with the same 25,000 challenges, but with 3 reads per challenge, allowing the measurement of inter-die uniqueness. The primary dataset was further filtered to remove the unstable CRPs. This filtered data was used purely to train the ML models, and nothing else.

D. Metric Definitions

Inter-Challenge HD quantifies the change in response with respect to the change in the challenge on a single board. For c distinct challenges each producing an L-bit response:

$$\text{Inter-Challenge HD} = \frac{2}{c(c-1)} \sum_{i=1}^{c-1} \sum_{j=i+1}^{c} \frac{\text{HD}(R_i, R_j)}{L} \times 100\%$$

where R_i and R_j are the responses to distinct challenges i and j. The ideal value is 50%, indicating statistically independent responses across the challenge space.

Intra-Die HD quantifies the change in response with respect to the same challenge across a single FPGA board. For a single challenge evaluated e times:

$$\text{Intra-HD} = \frac{2}{e(e-1)} \sum \sum \left(\frac{\text{HD}(R_i, R_j)}{L} \right) \times 100\%$$

where R_i and R_j represent the i-th and j-th physical evaluations of the same challenge. The ideal value is 0%, indicating a perfectly stable response.

Inter-Die HD quantifies the change in response with respect to the same challenge across multiple FPGA boards. For k distinct FPGA boards:

$$\text{Inter-Die HD} = \frac{2}{k(k-1)} \sum \sum \left(\frac{\text{HD}(R_i, R_j)}{L} \right) \times 100\%$$

where R_i and R_j are the responses from board i and board j. The ideal value is 50%, indicating that the PUF cannot be cloned on another board.

IV. Results and Analysis

All metrics were computed offline from the extracted multi-board datasets using a custom Python analysis pipeline.

A. Hardware Utilization

Table I reports the resource consumption of the Mirage PUF core on the XC7A100T, excluding the UART test infrastructure.

Table I. Post-Implementation Resource Utilization on Artix-7 XC7A100T

Resource	Used	Available	Utilization
Slice LUTs (SR latches, mask logic, PDL)	32	63,400	<0.05%
Slice Registers (FFs)	10	126,800	<0.01%
CARRY4 Primitives	64	15,850	0.40%
Slice LUTs (FSM + LFSR wrapper)	186	63,400	0.29%

The core PUF macro consumes under 0.5% of total device resources. Moreover, general-purpose LUT consumption is negligible.

B. Inter-Die Hamming Distance

The datasets, when evaluated, yielded a maximum Inter-Die HD of 13.04%. This constitutes a failure of unclonability under the standard PUF security model. This failure, upon a closer look at existing literature, seems to stem from the intrinsic characteristics of the CARRY4 primitives, which are optimized by the foundry for timing closure and yield. This leaves insufficient variation for the PUF to be truly unique across boards. There are architectures that achieve near-ideal Inter-Die HD [7, 10, 11]. However, they do so by routing delay paths through general-purpose interconnect, which limits reproducibility, or by using architectures that amplify small variations into stable binary distinctions, which increases area required and the noise floor.

C. Inter-Challenge Hamming Distance

The Mirage PUF achieved an Inter-Challenge HD of 49.99%. This shows that distinct challenges produce responses that are statistically indistinguishable random. This is also partly a reason for the architecture's ML resistance, as is reported in following sections.

D. Intra-Die Hamming Distance

The Mirage PUF exhibited an Intra-HD of 1.37%. This is well below the 5% threshold commonly cited in the PUF literature [1] and is achieved without any post-processing.

E. CRP Stability Distribution

Not all 25,000 CRPs are equally stable. Fig. 2 profiles the percentage of the key space meeting a given bit-error tolerance across the entire dataset. Permitting a 3-bit tolerance recovers 85.02% of the entire 2^{112} challenge space. This provides approximately 4.4 x 10^{33} stable CRPs. For contrast, an edge device will likely only use a few thousand CRPs in its lifetime.

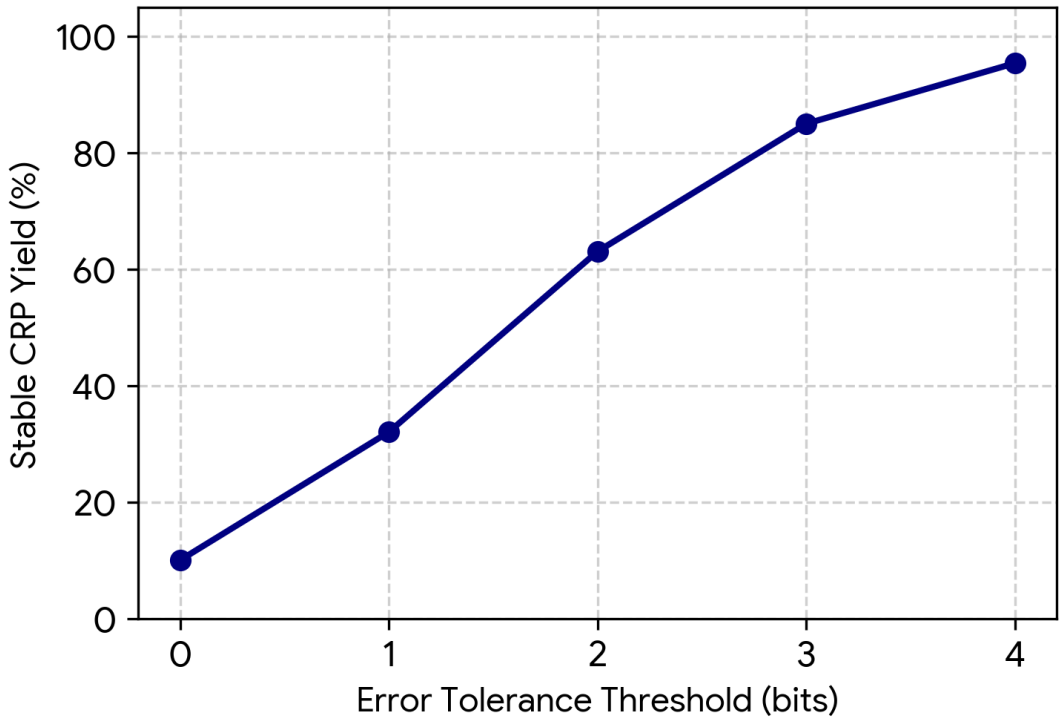


Fig. 2. Stable CRP yield as a function of error tolerance threshold. A 3-bit server-side tolerance recovers 85.02% of the key space without requiring on-chip ECC.

F. Resistance to Machine Learning Modeling Attacks

To give the ML models the best chance at cracking the PUF, they were given a filtered dataset containing stabilized CRPs exclusively, which came out to approximately 21,000 CRPs.

To ensure rigorous evaluation, both white-box and black-box attacks were simulated. For white-box attacks (like logistic regression and MLPs), the models were given full internal information, including all 16 possible mutated challenges from the masking process. Even so, prediction accuracy remained near the 50% random-chance baseline. As shown in Table II, the maximum recorded accuracy was 56.20%, achieved by LightGBM at 16,000 training CRPs. Random Forest was the only model that showed a monotonic increase in accuracy. However, even then, the accuracy improved by only 3% when the training dataset was expanded from 500 CRPs to 16,000 CRPs. To reach a reliable 90% accuracy, the training

dataset might need to be a few million CRPs big, which is unrealistic, even for attackers with physical access.

Table II. ML Attack Prediction Accuracy vs. Training Dataset Size (noiseless CRPs)

Attack Model	500	1000	2000	5000	10000	16000	Max
Logistic Regression	49.78%	49.00%	49.94%	49.44%	50.06%	49.76%	50.06%
Shallow MLP	50.32%	50.24%	50.94%	52.08%	50.86%	50.86%	52.08%
Deep MLP	50.50%	50.78%	51.16%	51.92%	51.46%	51.64%	51.92%
Random Forest	50.76%	50.88%	51.44%	51.96%	52.24%	53.76%	53.76%
XG Boost	50.90%	52.54%	52.26%	54.62%	54.04%	55.28%	55.28%
Light GBM	51.14%	51.88%	54.20%	55.76%	55.00%	56.20%	56.20%

G. Comparison with State-of-the-Art

Table III situates the Mirage PUF against recent FPGA-based PUF designs across five metrics relevant to edge deployments. As discussed earlier, the Mirage PUF's inter-die HD is well below the ideal of 50%, which the other architectures have been able to reach with a small error margin. However, the Mirage PUF performs exceedingly well in all other metrics, including the core objective of ML model resistance.

Table III. Comparison with State-of-the-Art FPGA PUF Architectures.

Architecture	Platform	Inter-Die HD	Inter-Challenge HD	Intra-HD	Max ML Accuracy	Core Footprint
PCS-APUF [10]	Artix-7	51.34%	N/A	2.43%	>90% (ANN)	389 LUTs
FC-MA PUF [7]	Artix-7	49.68%	N/A	0.018%	52.33%	N/A
DDQ-APUF [11]	Artix-7	47.28%	N/A	0.05%	90% @ 27k CRPs	N/A
C4TERO [3]	Artix-7	49.08%	N/A	0.046%	N/A	535 LUTs, 256 C4
Mirage PUF (Proposed)	Artix-7	13.04%	49.99%	1.37%	56.20%	32 LUTs, 64 C4

V. Conclusion

The Mirage PUF is presented as both a secure hardware design and a study of how CARRY4-based delays behave on Artix-7 chips.

The key idea is the "avalanche mask," where four SR-latch arbiters read intermediate hidden internal signals from another delay chain. This mask is used to mutate the primary PUF, making it hard to predict. Because of this, different machine learning models—like logistic regression, random forests, gradient boosting, and neural networks—can only reach about 56% accuracy, even after training on a large dataset.

At the same time, testing across different FPGA boards shows a limitation: the outputs are not very different between chips (about 13%). This happens because CARRY4 routing is highly optimized and reduces natural variation, which is needed to make each device unique.

The results in this work provide a solid starting point for building such improved designs. Future work includes combining the Mirage PUF with another architecture to form a hybrid design that solves the Mirage PUF's problem of die-uniqueness, testing on other hardware platforms and checking performance under different voltage and temperature conditions.

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